



Dimensionality Reduction (Part 2 of 2): Nonlinear Methods

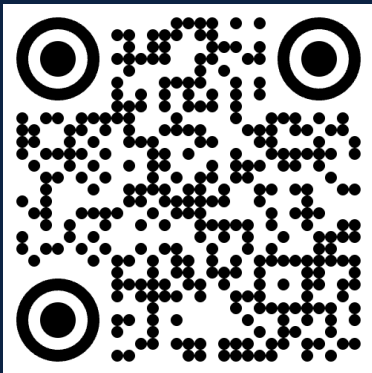
The Advanced R Series: Clustering & Classification

Slides and code created by Jesse Ghashti

March 30, 2026

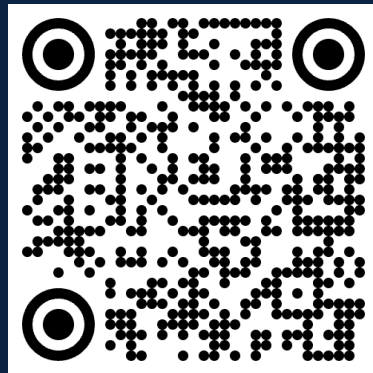
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←
Check out our other
CSC workshops!

GitHub code and
slides for today's
workshop (and pre-
vious workshops)



Alternatively, code/slides available at the bottom of
<https://csc-ubc-okanagan.github.io/workshops/>

Workshop Series Overview



Session	Topic	Date/Time
1	Hierarchical Models	Jan 14, 12:00 PM
2	Centroid-Based Models	Jan 19, 3:30 PM
3	Fuzzy Clustering	Jan 28, 1:00 PM
4	Distribution-Based Models	Feb 2, 3:30 PM
5	Density-Based Models	Feb 9, 3:30 PM
6	Graph-Based Models	Feb 23, 3:30 PM
7	Mixed-Type Data Quantification	Mar 4, 1:00 PM
8	Mixed-Type Data Clustering	Mar 11, 1:00 PM
9	Bias Reduction and Fairness	Mar 18, 1:00 PM
10	Dimensionality Reduction 1 of 2	Mar 23, 3:30 PM
11	Dimensionality Reduction 2 of 2	Mar 30, 3:30 PM

What we learned...

- Curse of dimensionality: distances become uniform in high dimensions
- PCA: find orthogonal directions of maximum variance
- Factor Analysis: model observed variables as driven by latent factors
- Applied both to UCI Communities and Crime data
- Reduced 98 features to 8 PCs while maintaining clustering quality
- PC₁ captured socioeconomic gradient; strongly correlated with crime

Today -> Nonlinear Dimensionality Reduction



Today we will...

- Understand why linear methods fail on nonlinear structure
- Learn t-SNE: preserve local neighborhoods via probability distributions
- Learn UMAP: faster method preserving both local and global structure
- Explore hyperparameter effects: perplexity, n_neighbors, min_dist
- Apply both methods to MNIST digits and Communities & Crime
- Compare linear vs nonlinear methods for clustering and visualization

Today we require...

```
library('ggplot2')  
library('dplyr')  
library('Rtsne')  
library('umap')  
library('mclust')  
library('keras3')
```

Linear methods assume linear relationships (of course)

Limitations of PCA/FA

- PCA finds linear combinations that maximize variance
- Data lying on curved manifolds cannot be “unfolded”
- Local neighbourhood structure may be destroyed by linear projection
- Global variance maximization may not capture cluster structure

What we want instead

- Preserve *local neighbourhoods*: nearby points stay nearby
- Allow nonlinear transformations to “unfold” manifolds
- Capture cluster structure even when clusters lie on complex surfaces

t-SNE

t-Distributed Stochastic Neighbor Embedding

- Preserve local neighborhoods
- Gaussian kernel in high-D
- t-distribution in low-D
- Minimize KL divergence
- Primarily for *visualization*

UMAP

Uniform Manifold Approximation & Projection

- Fuzzy topological structure
- k-nearest neighbor graph
- Cross-entropy optimization
- Faster than t-SNE
- Better global structure

Great for Dimensionality Reduction

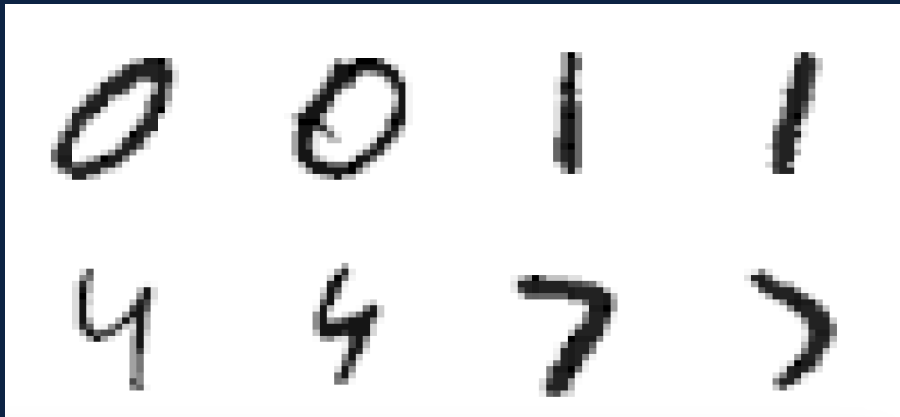
Data

- 28×28 grayscale images of handwritten digits (0–9)
- Each image = 784 pixels = 784 features
- We use a subset (only digits 0, 1, 4, 7)
- We use 500 samples per digit = 2000 total observations

Why MNIST?

- High-dimensional ($p = 784$) but with clear cluster structure
- Visually intuitive — we can see if reduction makes sense
- Nonlinear structure! Since digit shapes lie on curved manifolds

Sample MNIST Digits



Digits 0, 1, 4, 7 are visually distinct. The challenge: can dimensionality reduction recover this structure from 784 raw pixel values?

Step 1: High-dimensional similarities (Gaussian)

$$P_{j|i} = \frac{\exp(-\|\mathbf{x}_i - \mathbf{x}_j\|^2 / 2\sigma_i^2)}{\sum_{k \neq i} \exp(-\|\mathbf{x}_i - \mathbf{x}_k\|^2 / 2\sigma_i^2)}$$

Symmetrize: $P_{ij} = (P_{j|i} + P_{i|j}) / 2n$

Step 2: Low-dimensional similarities (Student-t)

$$Q_{ij} = \frac{(1 + \|y_i - y_j\|^2)^{-1}}{\sum_{k \neq l} (1 + \|y_k - y_l\|^2)^{-1}}$$

Step 3: Minimize KL divergence

$$\text{Cost} = \sum_{i \neq j} P_{ij} \log \frac{P_{ij}}{Q_{ij}}$$

The Crowding Problem

In high dimensions, a point has many “moderately close” neighbours. When projected to 2D, these all compete for limited space around each point.

So we use a heavy-tailed distribution in low-D

- t-distribution has heavier tails than Gaussian
- Dissimilar points can be placed *far apart* without huge cost
- Similar points are still attracted together

The t-distribution is important for the t-SNE to work for visualization — it solves the crowding problem that was a limitation of earlier methods (SNE).

Perplexity controls the effective number of neighbours

$$\text{Perplexity} = 2^{H(P_i)} \quad \text{where } H(P_i) = - \sum_j P_{j|i} \log_2 P_{j|i}$$

- Bandwidth σ_i is set so each point has specified perplexity
- Low perplexity (5–10): focus on very local structure
- High perplexity (30–50): consider broader neighbourhood

Rule of thumb: perplexity should be smaller than $n/3$


```
library(Rtsne)
set.seed(2026)
tsneMnist <- Rtsne(mnistScaled, dims = 2, perplexity = 30, max_iter = 1000, check_duplicates = FALSE)
tsneCoords <- tsneMnist$Y
```

Parameter	Description
dims	Output dimensionality (usually 2 or 3)
perplexity	Effective number of neighbours (5–50)
max_iter	Number of optimization iterations
theta	Speed/accuracy tradeoff (0 = exact, default 0.5)

Step 1: Construct fuzzy simplicial complex

- Build k-nearest neighbour graph
- Assign edge weights based on distance: $w_{ij} = \exp(-(d_{ij} - \rho_i)/\sigma_i)$
- ρ_i = distance to nearest neighbour (local connectivity)
- Symmetrize: $w_{ij}^{sym} = w_{ij} + w_{ji} - w_{ij} \cdot w_{ji}$

Step 2: Optimize low-dimensional layout

Minimize cross-entropy between high-D and low-D fuzzy sets:

$$\sum_{ij} \left[w_{ij} \log \frac{w_{ij}}{v_{ij}} + (1 - w_{ij}) \log \frac{1 - w_{ij}}{1 - v_{ij}} \right]$$

n_neighbors

Controls local vs global balance

- Smaller (5–15): tight local clusters
- Larger (30–100): more global structure
- Similar role to perplexity in t-SNE

min_dist

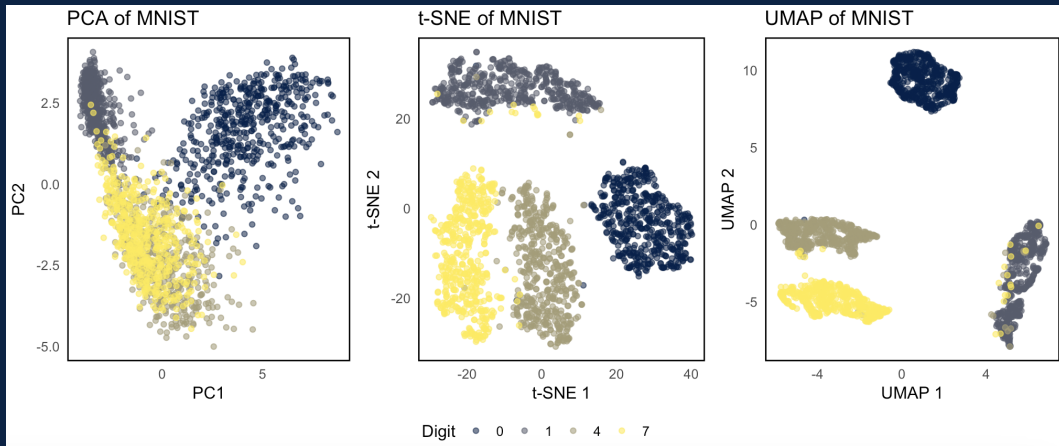
Controls cluster tightness

- Small (0–0.1): tight, dense clusters
- Large (0.5–1): spread out
- Affects visual appearance

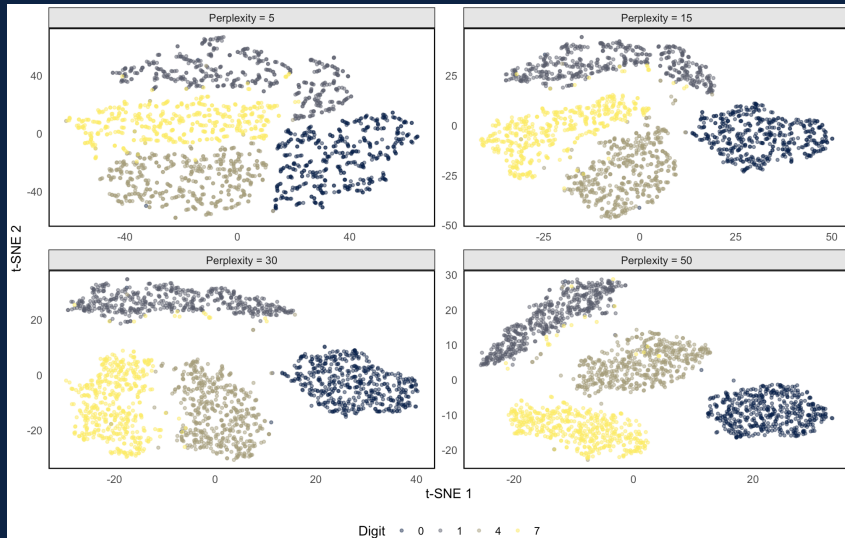
```
library(umap)
umapConfig <- umap.defaults
umapConfig$n_neighbors <- 15
umapConfig$min_dist <- 0.1
umapMnist <- umap(mnistScaled, config = umapConfig)
umapCoords <- umapMnist$layout
```

Aspect	t-SNE	UMAP
Speed	Slower	Faster
Global structure	Weak preservation	Better preservation
Reproducibility	Sensitive to initialization	More stable
New data	Cannot embed new points	Can embed new points
Theory	Probabilistic (KL divergence)	Topological (Riemannian)

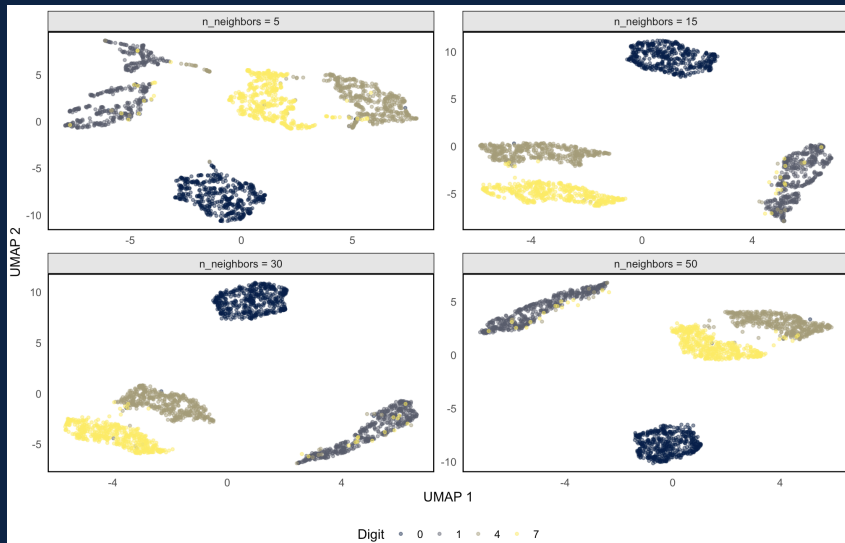
MNIST -> PCA vs t-SNE vs UMAP



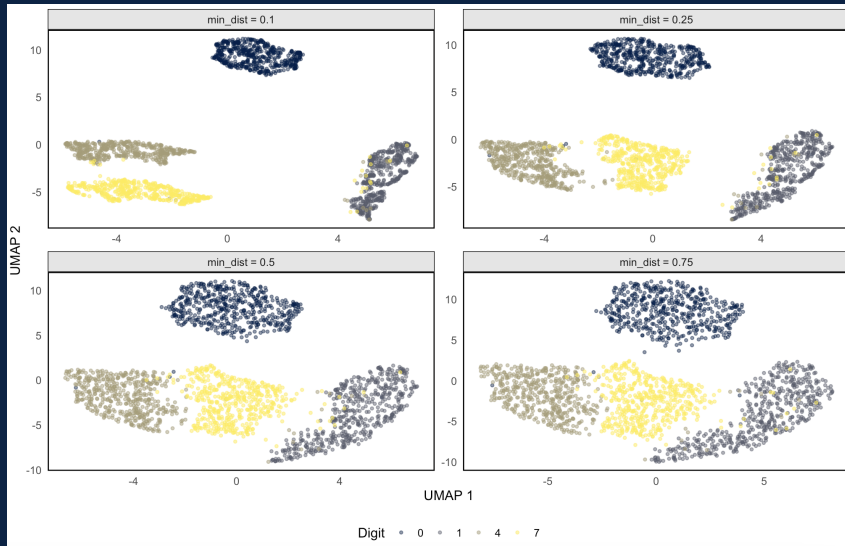
Effect of Perplexity on t-SNE (MNIST)



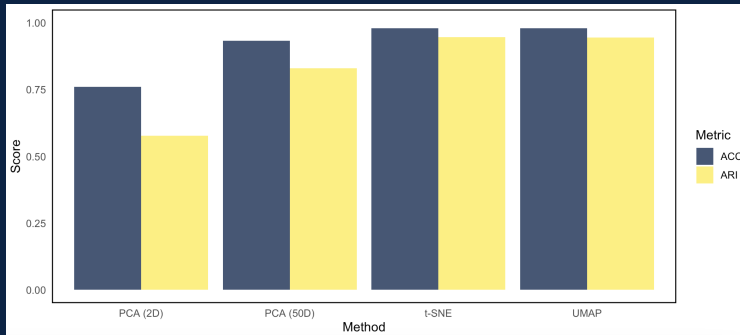
Effect of n_neighbors on UMAP (MNIST)



Effect of min_dist on UMAP (MNIST)

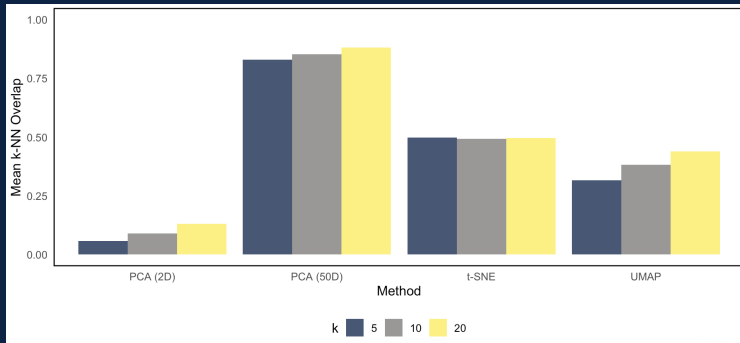


MNIST -> Clustering Performance



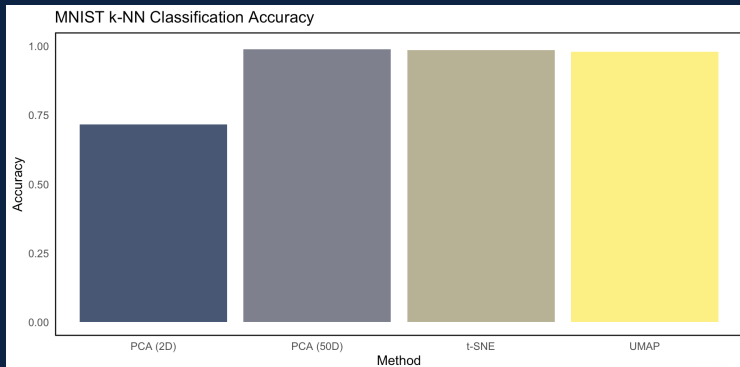
Nonlinear methods (t-SNE, UMAP) outperform PCA for clustering handwritten digits. The 2D nonlinear embeddings capture cluster structure that PCA misses even with 50 dimensions.

MNIST -> Neighborhood Preservation



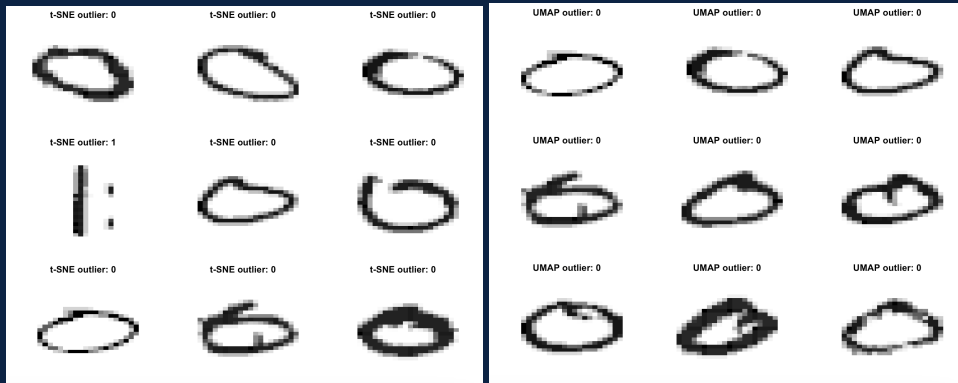
k-NN Overlap: fraction of original k nearest neighbours preserved in the embedding. Higher is better. t-SNE and UMAP preserve local structure much better than PCA.

MNIST -> k-NN Classification in Reduced Space



When using the embedding for downstream k-NN classification, nonlinear methods provide better class separation in 2D than PCA with 50D.

MNIST -> Outlier Analysis



Outliers in the embedding often correspond to ambiguous or atypical digit samples.

Same data as Workshop 10

- 1994 US communities, 98 socioeconomic features
- Binary labels: HighCrime vs LowCrime (65th percentile threshold)
- Features normalized to $[0, 1]$

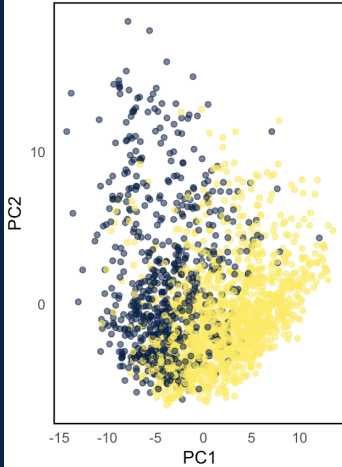
Questions

- Do nonlinear methods improve clustering performance?
- How do t-SNE and UMAP compare to PCA on tabular data?
- Is structure linear or nonlinear in this dataset?

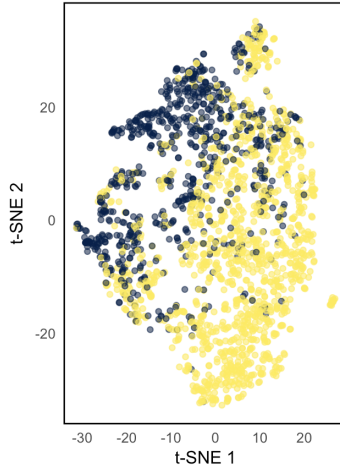
Crime Data -> PCA vs t-SNE vs UMAP



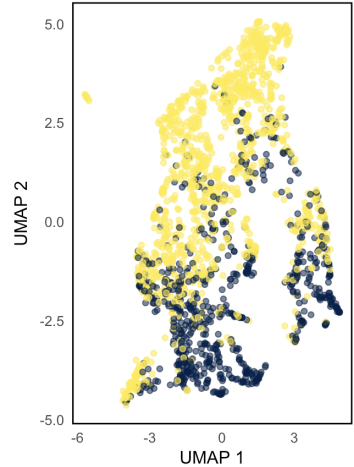
PCA of Communities



t-SNE of Communities

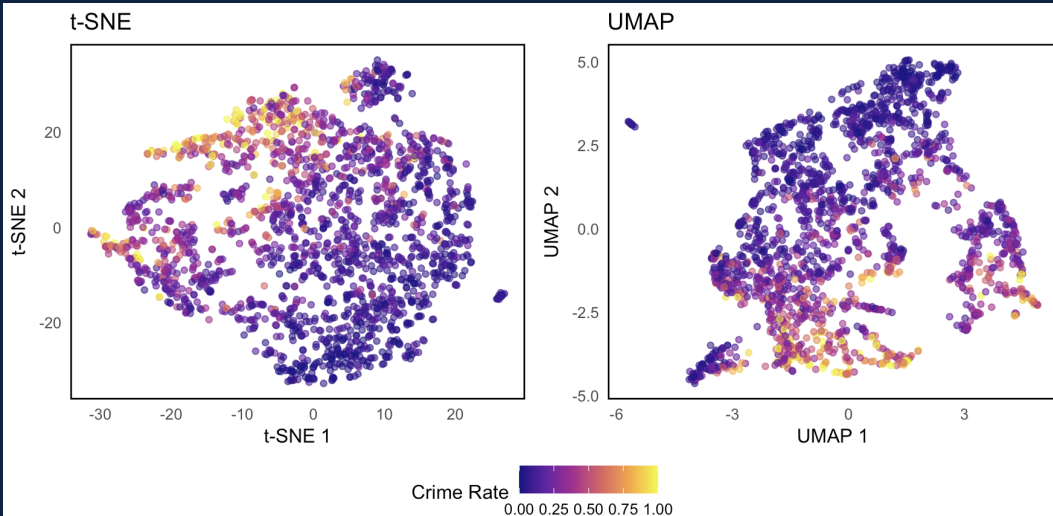


UMAP of Communities

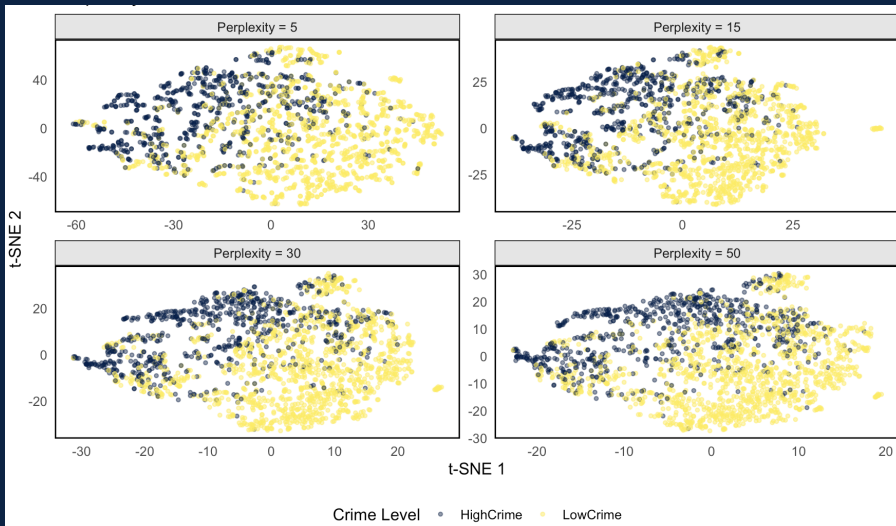


Crime Level ● HighCrime ● LowCrime

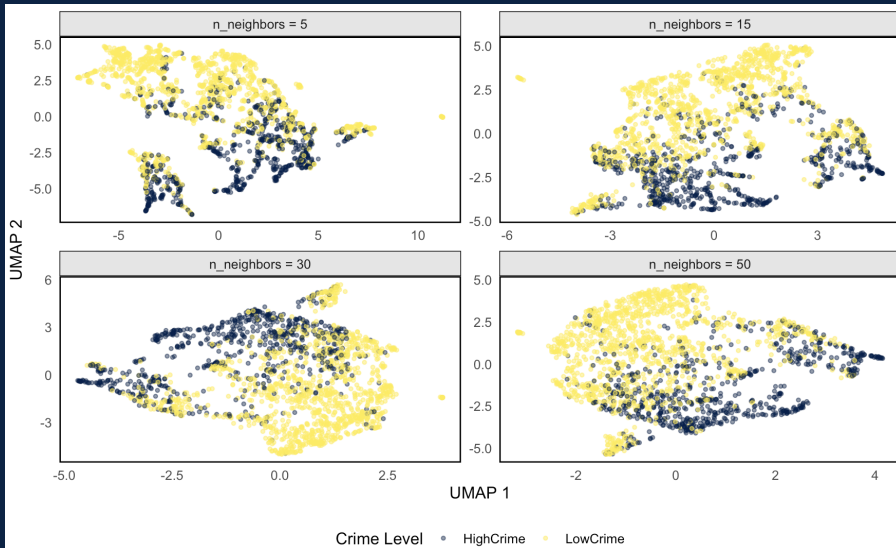
Crime Data -> Continuous Crime Rate



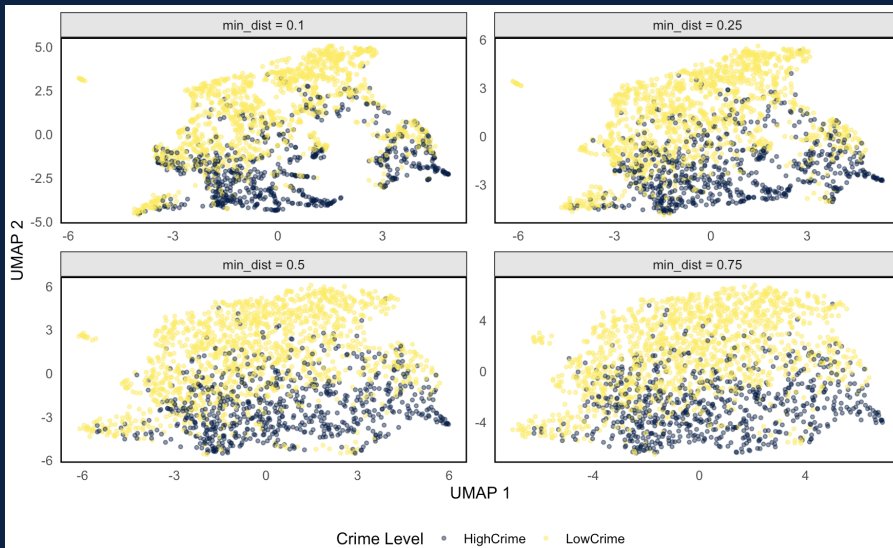
Effect of Perplexity on Crime Data (t-SNE)



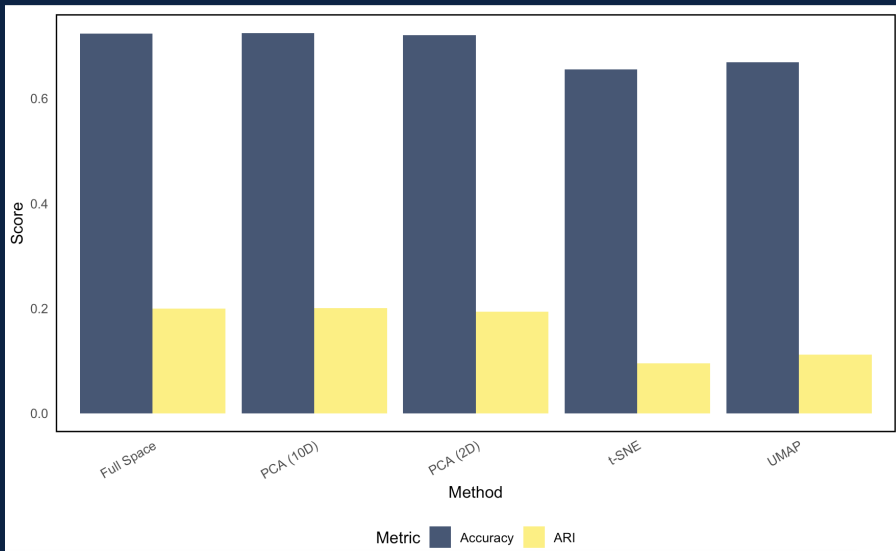
Effect of n_neighbors on Crime Data (UMAP)



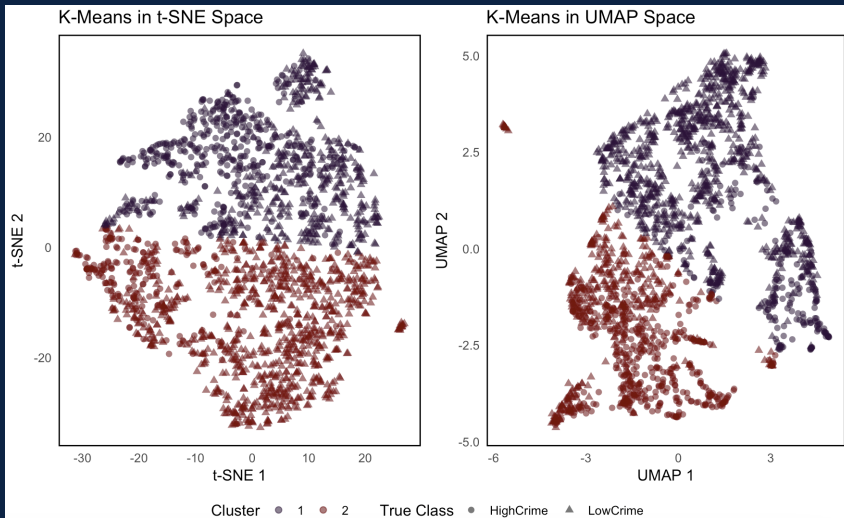
Effect of min_dist on Crime Data (UMAP)



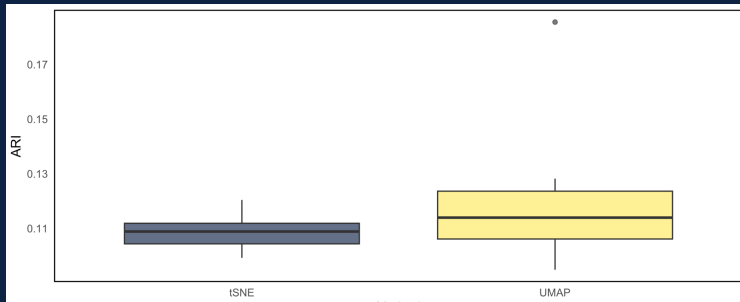
Crime Data -> Clustering Performance



Crime Data -> K-Means in Reduced Spaces

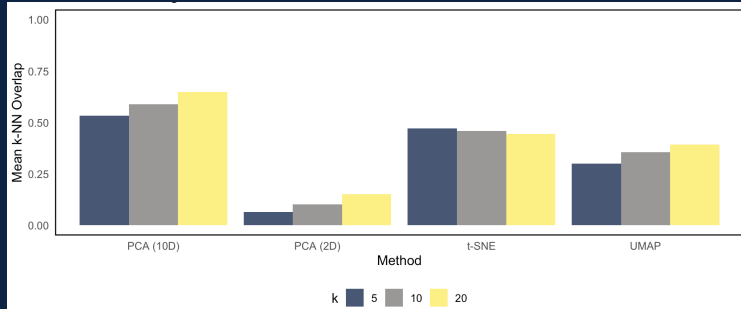


Stability Analysis -> t-SNE vs UMAP

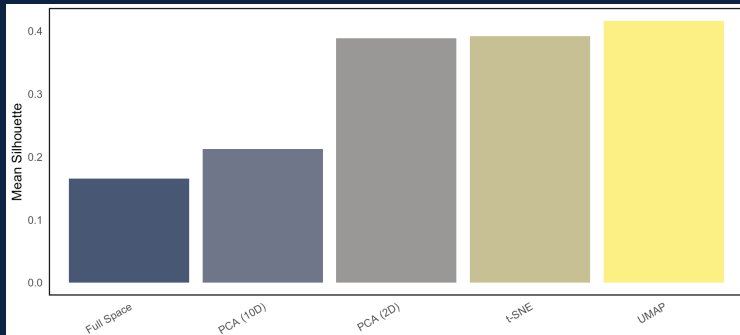


Consider running multiple seeds and checking stability.

Crime Data -> Neighborhood Preservation

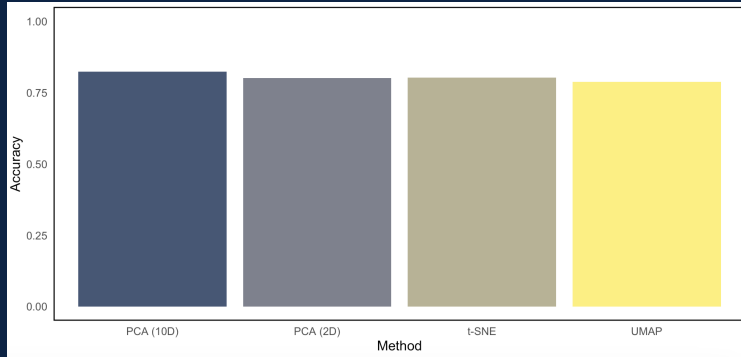


Crime Data -> Silhouette Scores

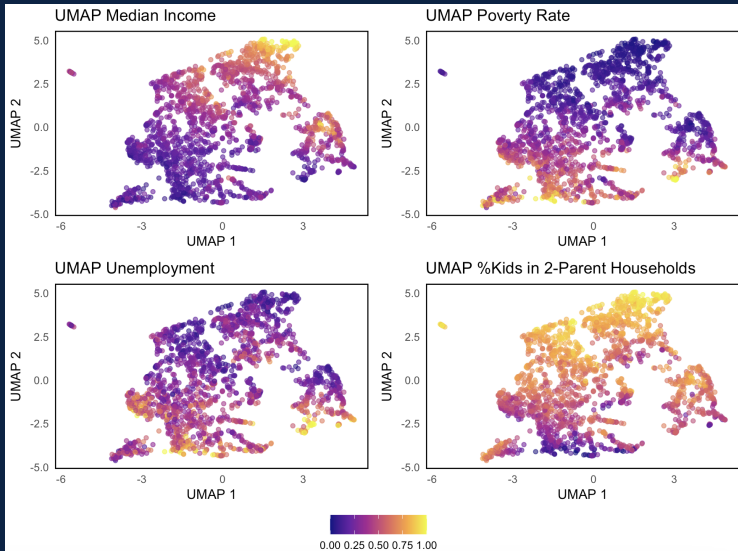


Silhouette scores measure cluster compactness and separation. Higher is better. Nonlinear methods produce better-separated clusters in the embedding space.

Crime Data -> k-NN Classification Accuracy



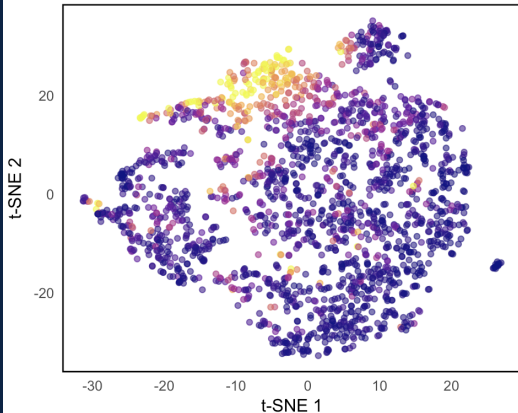
Crime Data -> Overlays on UMAP



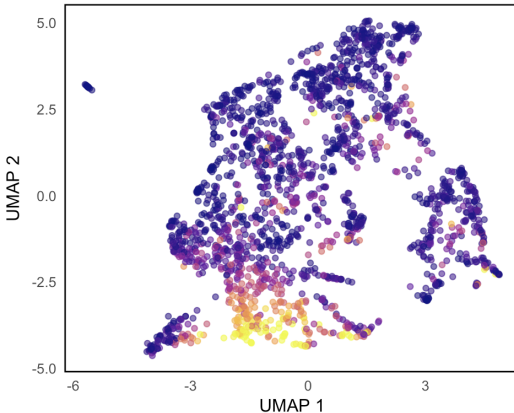
Fairness Revisited -> Race in Embeddings



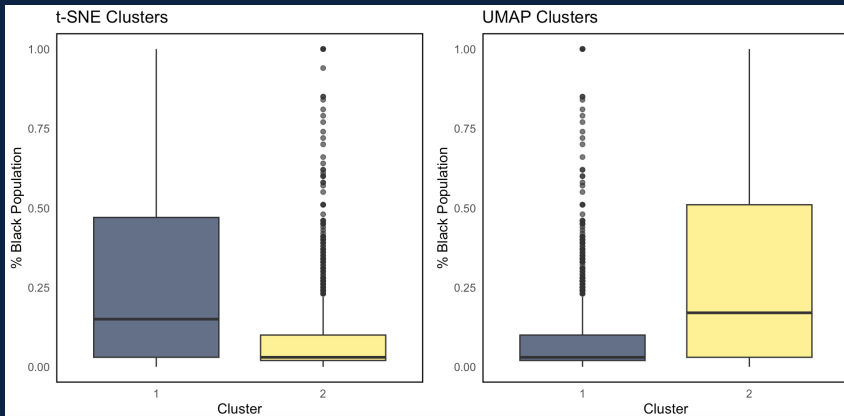
t-SNE: Racial Composition



UMAP: Racial Composition



Racial Composition by Cluster



As with PCA, nonlinear embeddings also encode racial structure. Clusters have different racial compositions. Fairness methods from Workshop 9 could be applied post-reduction.

Things to watch out for

- Cluster sizes are not meaningful: t-SNE/UMAP distort density
- Distances between clusters are not meaningful: only *within*-cluster distances
- Hyperparameters matter: always try multiple values
- Stochastic: run multiple times, check stability
- Not for inference: visualization only, not statistical testing

If you see clusters in t-SNE, they might be real. If you don't see clusters, that doesn't mean they don't exist.

- Linear methods (PCA, FA) cannot unfold nonlinear manifolds
- t-SNE preserves local neighbourhoods via Gaussian/t-distribution mismatch
- UMAP is faster, preserves more global structure, based on topology
- Both are great at visualization
- Hyperparameters (perplexity, n_neighbors, min_dist) control local/global tradeoff
- On MNIST: huge improvement over PCA for clustering
- On Crime data: maybe we need to tune some parameters better, or maybe the structure is more linear!

Additional Questions?
Book an Appointment!



What we covered:

- > Hierarchical, centroid-based, fuzzy, distribution-based, density-based, and graph-based clustering
- > Mixed-type data handling
- > Bias reduction and fairness
- > Linear and nonlinear dimensionality reduction

What else could you explore? Deep learning methods (autoencoders), spectral methods (Laplacian Eigenmaps), other manifold learning (Isomap, LLE)

Thank You!

Questions?

Workshop Materials:

<https://csc-ubc-okanagan.github.io/workshops/>

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